

[Titel]

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Abstract

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1 Introduction

Algorithms are increasingly involved in consequential decisions. Many of these decisions carry consequences that fall, at least in part, on others. Physicians rely on diagnostic systems when treating patients, firms let screening software rank job applicants, and financial advisors draw on algorithmic tools when managing client portfolios. In each case a human chooses whether to put the algorithm in charge. Algorithm use thus becomes a delegation decision and with that, two questions arise. Does delegating a decision to an algorithm change the responsibility that others assign to the person who delegated it? And does the prospect of being held responsible motivate delegation? This paper provides experimental evidence on both.

For delegation between humans, the experimental literature points in a clear direction. Delegation shifts responsibility. Principals who delegate an unpopular decision to a human intermediary are punished less for it, feel less responsible for it, and delegate strategically for precisely this reason (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Recent evidence suggests that algorithms possibly inherit the intermediary’s role. Delegation to an algorithm increases dishonesty by obscuring intent (Köbis et al., 2025), is used to extract surplus while shedding moral responsibility (Tontrup and Sprigman, 2025), and becomes more attractive when the moral stakes of the decision are real (Hüholt and Szech, 2026). Accordingly, decision-makers who face potential punishment should be drawn to the algorithm, and delegation should protect them when outcomes turn out badly. Yet no existing design varies whether punishment is possible, so whether anticipated punishment motivates delegation to an algorithm has not been tested directly.

We test both predictions in a preregistered online experiment with 322 participants.¹ A decision-maker, Player A, faces a prediction task (predicting a person’s weight from a facial image and the person’s height) with monetary consequences for a matched recipient, Player B. Player A has the option to either make the consequential prediction herself or delegate it to an algorithm that has been calibrated to perform equally well at the task. The environment we study, a prediction under genuine uncertainty made by a decision-maker with no material stake in the affected party’s outcome, resembles the settings in which algorithmic delegation typically operates in practice. We then vary between subjects whether Player B can reduce Player A’s earnings: In the *Punishment* condition, we elicit Player B’s punishment for every combination of delegation choice and outcome, providing a within-subject test of whether delegation insulates Player A from punishment, conditional on the outcome. In the *No-Punishment* condition, no such punishment is possible. Comparing delegation rates across the two conditions turns our motivational question into a testable treatment comparison. With everything except the possibility of punishment held fixed, any difference in delegation is attributable to its prospect.

First, when punishment is possible, 42.5 percent of decision-makers delegate to the algorithm. When it is not, 59.3 percent do. Thus, we find that the prospect of punishment significantly

¹The pre-registration (AsPredicted #133980) is available at <https://aspredicted.org/hs9az8.pdf> and reproduced in full in Appendix C.

reduces delegation shares by roughly 17 percentage points. Second, delegation offers no protection against punishment. Conditional on the outcome, the punishment of delegated and self-made decisions is statistically indistinguishable. Instead, punishment responds almost entirely to the quality of the outcome of the decision, not to how the decision was made. Belief evidence suggests that decision-makers anticipate this pattern correctly.

Thus, the possibility of punishment moves the delegation decision even though no punishment attaches to delegation. We examine several candidate explanations. A first candidate is anticipated punishment. Player A may delegate less because she expects delegated decisions to be punished more harshly. Her elicited beliefs speak against this story. On average, her expected punishment is no higher for delegated than for self-made decisions, matching how Player B actually punishes. A second candidate operates via effort. Facing potential punishment, Player A may keep the decision in the hope of outperforming the algorithm. If so, non-delegators compared to delegators should improve on their earlier performance precisely when punishment is possible. Yet, actual performance improvement concentrates where punishment is impossible. What the data do show is a selection pattern. When punishment is possible, it is disproportionately the better-performing Player As who keep the decision for themselves. A preference for producing a good outcome oneself and a distaste for delegating uncertain decisions are both consistent with this pattern. However, both are treatment-independent in that they should operate in both conditions alike and thus predict no difference in delegation rates. Lastly, not delegating may carry signaling value. By keeping the decision, Player A may aim to convey to Player B that she intends to try especially hard on his behalf, whereas delegating could be perceived as shirking the decision. Since there is some inherent task uncertainty, sending this message does not require actually trying harder. The choice itself is the signal, so the motive is consistent with the absence of measurable extra effort among non-delegators. And unlike the preference explanations, it is inherently treatment-dependent. The signal is worth sending only when Player B holds sanctioning power, which could generate the delegation gap we observe.

Our contribution is twofold. First, we provide the first direct test of whether the prospect of punishment motivates delegation to an algorithm, the motive the responsibility-avoidance literature identifies for human intermediaries (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Our environment departs from these canonical designs in three respects. The decision involves genuine uncertainty, so a bad outcome may reflect bad luck rather than bad intent. Its consequences fall entirely on the recipient, so there is no self-serving choice to police. And the intermediary is an algorithm rather than a person. In this environment, the motive operates with the opposite sign. Accountability disciplines algorithm use rather than fueling it, and it does so selectively among the decision-makers most likely to succeed. This rejection of the standard prediction in favor of the opposite direction is the paper’s main empirical contribution, and we view it as a behavioral counterpart to the regulatory intuition that holding humans accountable for algorithm-mediated decisions disciplines their use of those algorithms (European Union, 2024). The reversal also adds a potential determinant of algorithm use that operates through the decision context rather than through

the algorithm itself. Our results are consistent with decision-makers keeping the decision because delegating it may look like shirking to the person it affects, even when the algorithm performs equally well. Second, we document how affected parties assign responsibility for algorithmic delegation when the delegation choice is uninformative about ability. Punishment then tracks realized outcomes, and the insulation that delegation appeared to offer in earlier designs disappears.

Our design most closely relates to [Feier et al. \(2022\)](#), who find that decision-makers are held less responsible for outcomes delegated to an algorithm than for outcomes of their own actions. However, in their design each subject both delegates and evaluates the delegation decision, and under their session-level calibration of the algorithm, delegation is informative about and primarily driven by self-assessed relative ability. Subjects may then sanction perceived overconfidence rather than the delegation itself. Once these channels are closed, as in our design, the punishment differential disappears. Their treatments vary the type of delegate under a fixed punishment regime, whereas we hold the delegate fixed and vary the punishment regime. The two studies thus answer complementary questions, and ours isolates the role of the prospect of punishment.

The remainder of the paper is organized as follows. Section 2 relates the study to the literature. Section 3 presents the experimental design and hypotheses. Section 4 reports the results and examines candidate mechanisms. Section 5 concludes.

2 Related Literature

Responsibility avoidance through delegation

Delegation has long been studied as an efficiency-enhancing institutional arrangement with principals delegating to agents who have better information, lower opportunity costs, or specialized skills. Alongside these efficiency motives, the literature has identified a range of strategic motives for delegation, e.g. as a commitment device that alters the equilibrium ([Schelling, 1960](#); [Fershtman and Gneezy, 2001](#)), as a way to motivate the agent ([Aghion and Tirole, 1997](#)), or as a means of shifting responsibility for the outcome away from the principal, i.e. principals may delegate not because the agent is more competent or cheaper, but because doing so shifts responsibility for the resulting outcome away from themselves — both in their own eyes (felt responsibility for the outcome) and in the eyes of others (the blame others attribute to them). The political-science and public-choice literatures have long recognized that delegation can serve as a strategy for shifting blame onto an intermediary ([Weaver, 1986](#); [Fiorina, 1986](#)).

Experimental economics has established the same motive in incentivized settings, primarily using modified dictator games. [Bartling and Fischbacher \(2012\)](#) provide the canonical experimental evidence. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to another person whose monetary

incentives are aligned with the dictator's. A recipient, who is adversely affected by the unfair allocation, may then assign costly punishment to the dictator and the delegee. They find that through delegation punishment is effectively shifted from the dictator to the intermediary, and the prospect of avoiding punishment constitutes a strong motive for the delegation decision itself. [Coffman \(2011\)](#) reinterprets the mechanism behind this punishment reduction. He finds that the principal can no longer avoid punishment when she directly interacts with the receiver despite the intermediary's presence, suggesting that what reduces punishment is the absence of direct interaction with the affected party rather than any transfer of blame to a deciding intermediary. This holds even when the intermediary remains completely passive. [Oexl and Grossman \(2013\)](#) sharpen this reading further by modifying the canonical game so that delegation necessarily eliminates the fair option. The principal nevertheless escapes punishment, with the intermediary actively choosing between two unfair allocations after delegation. Finally, both [Bartling and Fischbacher \(2012\)](#) (with a die roll) and [Oexl and Grossman \(2013\)](#) (with an intermediary-triggered random draw) include treatments that substitute a randomization device for the human intermediary. In both, the principal still shifts some punishment to the device, though less than under delegation to a deciding human. Consistent with this, in [Bartling and Fischbacher \(2012\)](#) fewer dictators delegate to the die than delegate to a human delegee in the corresponding treatment. Two components of responsibility shifting thus emerge — the absence of direct interaction between principal and recipient drives the reduction in principal punishment, while the act of choosing, rather than any contribution to outcome unfairness, is what transfers blame to the intermediary.

The motive is not confined to settings with a punishment stage. [Hamman et al. \(2010\)](#) extend this picture in a setting without punishment opportunity. Still, principals' transfers to recipients fall sharply when they delegate, indicating that responsibility-shifting can operate through felt responsibility or moral wiggle room rather than through the avoidance of punishment. Qualitative self-reported responsibility and recipient-side blame attributions are consistent with this reading with dictators feeling less responsible when they delegate, and recipients placing less blame on principals than on intermediaries.

Beyond the canonical dictator-game setting, the responsibility-shifting motive, and the strategic use of delegation it gives rise to, has been documented across a range of domains. In incentivized ultimatum bargaining, proposers using a hired delegate extract larger surpluses ([Fershtman and Gneezy, 2001](#)). In sequential voting, non-pivotal voters are punished less than pivotal ones for the same group outcome and voters strategically vote against their preferred allocation to avoid being pivotal ([Bartling et al., 2015](#)). For acts of deception, principals pay an agent to lie on their behalf ([Erat, 2013](#)) and are more willing to lie through a delegate than directly ([Gawn and Innes, 2019](#); [Kandul and Kirchkamp, 2018](#)). In policy-making, observers rate delegating legislatures as less blameworthy than legislatures who decide directly, even when the delegate is powerless to change the outcome ([Hill, 2015](#)). [Freer et al. \(2024\)](#) isolate the motive in financial decision-making, finding that a substantial fraction of investors delegate even purely luck-based lottery choices, where alternative motives such as decision costs, performance-chasing, or risk

tolerance cannot rationalize the choice.

In sum, the canonical literature documents that in settings with self-interested choice, delegation reduces both the principal’s punishment and her felt responsibility, in turn motivating delegation. Our setting departs from this canonical design along three dimensions. First, the intermediary is algorithmic rather than human. Second, the outcome falls entirely on a passive recipient rather than partly on the principal herself. Third, the underlying decision involves genuine predictive uncertainty rather than a known fair–unfair choice, removing the transparent self-serving intent that punishment sanctions in the canonical designs. Whether the responsibility-shifting motive retains its force under these conditions is the question we take up.

Delegation to algorithms

A large and rapidly growing empirical literature studies when and why people are willing to use algorithmic decision aids. [Dietvorst et al. \(2015\)](#) introduce the term *algorithm aversion* — the finding that people prefer their own (or other humans’) judgement over algorithmic predictions, even after observing that algorithms outperform human decision-makers.² In contrast, [Logg et al. \(2019\)](#) document conditions under which subjects show *appreciation* of algorithmic advice over human advice. The picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. [Burton et al. \(2020\)](#), [Mahmud et al. \(2022\)](#), [Chugunova and Sele \(2022\)](#), [Qin et al. \(2025\)](#) and [Irlenbusch \(2026\)](#) provide systematic reviews of this literature and document a large array of moderators for when and why algorithms are used in decision-making.

For example, on the algorithm side, algorithm aversion is stronger when the algorithm cannot be observed to learn from its mistakes ([Berger et al., 2021](#)), and when the algorithm remains a black box to its users ([Litterscheidt and Streich, 2020](#)). Evidence about relative performance gathered through experience and feedback reduces aversion ([Filiz et al., 2021](#)), whereas explanations of how the algorithm works leave acceptance unchanged ([Dargnies et al., 2026](#)). On the task side, algorithm aversion rises with the degree of irreducible uncertainty ([Dietvorst and Bharti, 2020](#)), with the perceived subjectivity of the task ([Castelo et al., 2019](#)), while reliance on algorithmic advice increases with task difficulty ([Bogert et al., 2021](#)) and under time pressure ([Jung and Seiter, 2021](#)). On the control side, people use imperfect algorithms far more readily when they retain even a minimal ability to modify the output ([Dietvorst et al., 2018](#)). Likewise, experimental firms delegate pricing to an algorithm more often when they can override its decisions ([Normann et al., 2025](#)). Part of algorithm aversion may thus reflect a reluctance to relinquish decision authority as such rather than distrust of algorithms ([Ivanova-Stenzel and Tolksdorf, 2025](#)), and performance feedback raises reliance on algorithms without eliminating

²Interest in the comparison between human and statistical judgement is long-standing — see [Meehl \(1954\)](#) and [Dawes \(1979\)](#) for the classic demonstrations that simple statistical rules outperform clinical judgement and are resisted nonetheless.

the desire for personal control (Ivanova-Stenzel and Tolksdorf, 2024).

To do: Supplement with more recent evidence; Explain that while context, task, etc. may matter for whether shifting responsibility is possible and this motive may impact demand for algorithms, level-effects of a pure preference for algorithms are constant across treatments.

Responsibility perceptions of decisions involving algorithms

Within this body of work, a smaller subset focuses on algorithm use preferences in settings whose consequences fall partly or wholly on someone other than the decision-maker, sometimes called the *moral domain* (Gogoll and Uhl, 2018). Much of this work concludes that aversion to algorithms is amplified once third-party welfare is at stake. Gogoll and Uhl (2018) document this in an incentivized calculation task with performance uncertainty, Bigman and Gray (2018) in vignettes across high-stakes moral domains (driving, legal, medical, military), and Jauernig et al. (2022) identify *moral discretion* as the human capacity that people specifically value in such settings.³

Once decisions affect a third party, the responsibility-shifting motive established for human delegation becomes relevant for algorithmic delegation as well — and with it the question of how observers apportion blame when an algorithm is involved in the decision. [Comment: In principle, felt responsibility may be reducable by delegating also in decisions that only affect oneself.] A first strand of evidence, predominantly vignette-based and unincentivized, examines how third parties attribute responsibility and blame to human actors involved with algorithms. In the medical domain, Pezzo and Pezzo (2006) document that a physician who follows a (bad) recommendation from a computerised decision aid is faulted less than one who errs unaided. Tobia et al. (2021) extend the discussion to liability law, where physicians who follow standard-care AI recommendations face reduced legal exposure. The authors argue that this constitutes an incentive to follow such recommendations specifically because doing so shields physicians from blame. In hiring, algorithmic discrimination triggers less moral outrage than identical human discrimination (Bigman et al., 2023). In the workplace setting, Naquin and Kurtzberg (2004) find that observers attribute less responsibility to a company when an accident involves technology. Across multiple moral domains (driving, legal, medical, military), Bigman and Gray (2018) document a robust “algorithmic outrage asymmetry”, by which moral violations by algorithms produce less outrage than identical violations by humans. Shank et al. (2019) consider moral violations produced by human-algorithm pairs and find that humans who monitor algorithms are blamed less than humans acting alone or humans receiving algorithm recommendations, suggesting that responsibility attributions are sensitive to the precise division of authority between human and machine.

To do: Supplement/replace with more recent evidence.

³For recent surveys of this area see Bonnefon et al. (2024) and, with a legal-policy focus, Sunstein and Gaffe (2024).

Evidence using incentivized laboratory experiments is younger and smaller. [Kirchkamp and Strobel \(2019\)](#) compare a modified dictator game in which the dictator either acts alone or makes the decision jointly with a computer. They find no significant difference in the dictator’s self-felt responsibility between human-computer and human-human teams, although a tendency towards more selfish allocations under shared decision-making with humans. [Maasland and Weißmüller \(2022\)](#) find in a computer-aided HR setting that algorithm aversion *dominates* blame avoidance, i.e. the prospect of using an algorithm to delegate an unpleasant HR decision (a dismissal) does not overcome the underlying aversion to algorithms in this domain.

A more recent strand documents the opposite pattern, in which algorithmic intermediaries serve as moral cover. [Chevrier and Teixeira \(2024\)](#) extend the blame chain to include the programmer behind the algorithm. In their design, AI programmers and the delegator both escape punishment for inequalitarian AI-mediated allocations, and the resulting “moral wiggle room” is exploited. [Tontrup and Sprigman \(2025\)](#) examine delegation of a dictator-game allocation to ChatGPT and find that delegators transfer less to the recipient than non-delegators, indicating that AI delegation is used strategically to extract surplus while shedding moral responsibility, similarly to delegation to other humans in the canonical literature. [Hüholt and Szech \(2026\)](#) use an incentivized trolley-style donation choice between two real charitable allocations and find that subjects delegate the moral decision to an AI significantly more often than to a human counterpart. Across these three different settings, the AI-as-blame-shield motive is shared, and the direction of the effect on delegation is uniformly positive — in contrast to [Maasland and Weißmüller \(2022\)](#), where algorithm aversion dominates the responsibility-shifting motive in the same broad domain.

The most closely related study is [Feier et al. \(2022\)](#), who study delegation to an algorithm in an incentivized laboratory experiment. In their design, to determine a third party’s payoff subjects have the option to either rely on their own performance in a Raven-style logic puzzle or, depending on the treatment, delegate to another human or an algorithm. The subject’s own payoff is unaffected by this choice. Third parties may then reward or punish the principal. The headline finding is that decision-makers are held *less* responsible for decisions delegated to an algorithm than for decisions delegated to another human, conditional on outcome. Interestingly, in their experiment, punishment is only reduced significantly when delegating to an algorithm, but not to another human, compared to self-made decisions. Thus, algorithm-delegation, but not human-delegation, shields the principal from blame for bad outcomes.

Our design extends [Feier et al. \(2022\)](#) along three dimensions. First, where their algorithm is calibrated at the session level so that the delegation decision reflects beliefs about own ability *relative* to that distribution, in turn constituting a signal value about Player A’s self-perception that may contaminate downstream punishment as a measure of responsibility shifting per se, we calibrate the algorithm at the individual level, removing this relative-performance signal of delegation by construction.⁴ Second, where the same subject acts as both delegator and

⁴Indeed, their own analysis shows that the propensity to delegate is driven primarily by self-assessed errors in the logic task, not by the type of delegate. Punishment of a non-delegator may then partly sanction perceived

evaluator in their design, potentially introducing self-justification, hypocrisy aversion, and similar motives into the punishment decision, we separate the delegator and evaluator roles across disjoint subject pools.⁵ Third, their design does not include a treatment in which punishment is ruled out by construction, so that delegation rates between the two treatments may be different not because of differences in anticipated punishment, but more general preferences for algorithms. Thus, the motivational question of whether the prospect of punishment itself drives delegation is not identified. We add this manipulation directly, isolating the role of anticipated punishment in the delegation decision itself.

Thus, our experiment is, to our knowledge, the first to provide a clean test of blame-shifting onto an algorithm in a setting with inherent uncertainty, controlling for relative-performance concerns. It is also the first to vary the punishment possibility for an algorithmic delegate, and hence to test whether, in such a setting the prospect of punishment itself motivates the delegation decision.

3 Experimental Design

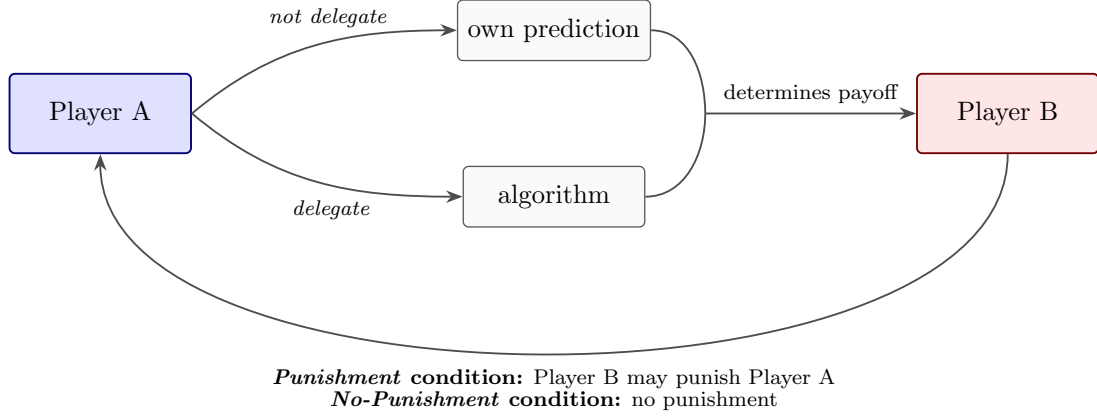
We ran an online experiment via the British platform Prolific (<https://www.prolific.com>) with UK participants. The experiment was programmed in oTree (Chen et al., 2016) and eligibility followed Prolific’s standard quality screens as well as a minimum 95% prior approval rate, a fluent-English filter and a desktop-only restriction (no mobile or tablet) to keep the screen layout consistent across subjects.

The design combines a between-subject manipulation of whether costly punishment is possible with a within-subject elicitation of conditional punishment behavior. Within each condition, participants assumed one of two roles. Player A decides whether to make a payoff-relevant prediction for Player B directly, or to delegate the prediction to an algorithm calibrated to match her own measured performance. Player B is the matched recipient whose payoff is fully determined by the outcome of Player A’s prediction, whether Player A makes it herself or delegates it to the algorithm. In the *Punishment* condition Player B may subsequently punish Player A, conditional on both the delegation decision and the realized outcome, while in the *No-Punishment* condition no punishment is possible. Figure 1 summarizes the basic setup.

overconfidence rather than responsibility-taking per se.

⁵For example, in their data, the evaluator’s own delegation decision is significantly correlated with the evaluation gap between delegated and non-delegated decisions after good outcomes ($r = -0.34$ in the machine treatment; own computations from their published data)

Figure 1: Overview of the experimental conditions



Procedure

Player A

After providing informed consent, Player A completed ten incentivized rounds of a visual prediction task. Specifically, in each round, she predicted a person’s weight from a facial image and the person’s height. Predictions could be entered via a slider or typed directly in kilograms, pounds, or stones-and-pounds.⁶ No feedback was provided between rounds. At the end of the experiment, one of the ten rounds was randomly selected for payment: Player A earned £4 if her prediction in that round was within 10 lbs of the true weight and £2 otherwise. These initial rounds served three purposes simultaneously: First, they established a stake from which any later punishment could be deducted. Second, they familiarized Player A with the task. Third, they generated the performance data needed to calibrate the algorithm.

The choice of a visual prediction task is not incidental. Algorithm preferences may depend sensitively on the algorithm’s perceived relative competence (Qin et al., 2025; Logg et al., 2019). Our design neutralizes this concern by calibrating the algorithm to match each subject’s individual performance. The visual nature of the task lends credibility to this equal-performance claim, as people have everyday familiarity with judging others’ build and appearance, and algorithmic error is plausible.⁷ In contrast, subjects may be skeptical that an algorithm could ever fail in purely mathematical tasks (Gogoll and Uhl, 2018; Feier et al., 2022). Finally, we chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations). This reflects the nature of many real-world applications of algorithms such as financial forecasting, medical diagnosis, fraud detection, in which outcomes are inherently uncertain and performance is evaluated ex post.

After the ten rounds, Player A was randomly paired with another participant, Player B, who had not completed the prediction task himself. Player A was informed that her final prediction would not affect her own payoff but would almost entirely determine Player B’s. A prediction

⁶All individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.

⁷Predicting body weight from an image has been used in earlier experiments (Logg et al., 2019).

within 10 lbs of the true weight would give Player B a high payoff (£5), and any less accurate prediction a low payoff (£1). Player A then learned that she could either make the prediction herself or delegate it to an algorithm. She was also told that, based on her first ten predictions, the algorithm would have performed equally well.

The algorithm was calibrated to mimic Player A’s own performance over the first ten rounds. Concretely, if Player A delegates the decision to the algorithm, the algorithm was equivalent to drawing a Bernoulli outcome with success probability equal to Player A’s empirical accuracy rate over her first ten rounds. For example, a subject who scored 3 out of 10 in the initial rounds thus delegates to an algorithm that produces the high payoff for Player B with 30% probability.

This individual-level calibration is a central design choice. Fixing the algorithm’s success probability to each subject’s own accuracy makes the equal-performance claim hold subject by subject, not merely on average across a pool of participants. This matters because beliefs about relative performance are otherwise likely a dominant driver of delegation.⁸ For example, under the session-level calibration of [Feier et al. \(2022\)](#), the delegation choice is largely driven by each subject’s sense of her ability relative to the algorithm. Matching the algorithm to each subject neutralizes this motive, and making this performance parity common knowledge prevents Player B from reading delegation as a signal of Player A’s believed relative competence. This closes off an important signal-extraction channel which may otherwise contribute to Player B’s punishment.

In the *Punishment* condition, Player A was further informed that Player B could reduce her payoff by up to £2, with the deduction depending on both her delegation decision and the realized outcome. In the *No-Punishment* condition, Player B had no such punishment option. The two delegation options (own decision / algorithm) were presented in random order across subjects. We attached no monetary cost to delegation. This brings us closer to Player A’s intrinsic preference over delegation in the *No-Punishment* condition, since with any positive cost, the responsibility-shifting motive would need to outweigh a price effect. Thus, costless delegation is standard in the literature ([Bartling and Fischbacher, 2012](#); [Coffman, 2011](#); [Feier et al., 2022](#)). If Player A chose to make the prediction herself, she entered her final prediction on the next screen. If she delegated instead, this screen was skipped.

We then asked Player A to report her beliefs about Player B’s punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). Beliefs were elicited on the same scale as actual punishment (£0–£2 in £0.10 increments) and the four scenarios were presented in block-wise random order. We chose not to incentivize belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivized mechanism could induce hedging behavior. Third, an incentive-compatible

⁸This motive would not inherently bias the treatment-effect comparison, since it should be balanced across the *Punishment* and *No-Punishment* conditions, but it might otherwise dominate the delegation decision and may obscure the responsibility-shifting motive of interest.

scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen. In the *No-Punishment* condition, no punishment was possible. Yet, we elicited beliefs about hypothetical punishment to maintain consistency across treatments.

Player A was then asked to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10. We elicited this belief after the delegation decision, and after Player A had been told that the algorithm would perform equally well. While learning about algorithm performance may lead her to revise her beliefs about her own performance, it is precisely this post-update belief that is the analytically relevant input for any subsequent behavior. The elicitation was incentivized: Player A earned an additional £0.30 with the probability she had assigned to her actual score.⁹ Player A was also asked to estimate the distribution of scores in the population, assigning to each score from 0 to 10 the share of participants she expected to have achieved it. Combined with her belief about her own score, this locates where she places herself in the distribution and yields a measure of perceived relative ability. One score was drawn at random at the end of the experiment, and Player A earned £0.30 if her estimate for it fell within five percentage points of the realized population share.

Finally, we elicited risk preferences using an incentivized multiple price list in the style of [Holt and Laury \(2002\)](#) and administered a short questionnaire on socio-demographic characteristics and technology affinity, before presenting Player A with her preliminary payoff. Figure 10 in Appendix B summarizes the experimental timeline for Player A.

Player B

After providing informed consent, Player B was told that he had been randomly matched with a Player A and was introduced to the task Player A had completed. He learned that his own payoff would be determined by the outcome of Player A’s final prediction - £5 if it fell within 10 lbs of the true weight and £1 otherwise - while Player A’s payoff depended solely on her performance in the initial ten rounds. Crucially, Player B was also told that Player A had the option to delegate the prediction to an algorithm that performs equally well as Player A had performed in the earlier rounds, and that Player A was aware of this, making it common knowledge.

In the *Punishment* condition, Player B could choose to reduce Player A’s payoff by up to £2 in £0.10 increments. We focus on punishment rather than reward because the responsibility-attribution literature has predominantly operationalized blame through punishment ([Bartling and Fischbacher, 2012](#); [Coffman, 2011](#); [Steffel et al., 2016](#)). Imposing strictly positive pun-

⁹While this linear scoring rule is not proper as its payoff-maximizing report places all mass on the single most likely score, irrespective of risk attitude, in practice most subjects spread probability mass across several scores.

ishment required Player B to forgo £0.10 of his own earnings, ensuring that punishment was costly to the punisher.¹⁰ We elicited punishment decisions for all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B), implementing only the decision matching the realized combination. Because the punishment decision is the central outcome variable for Player B, we included an attention check on the punishment-elicitation screen. In the *No-Punishment* condition, where actual punishment was not possible, we elicited hypothetical punishment decisions on the same scale to keep the experimental flow comparable across conditions. Player B’s perceptions of task difficulty, his risk preferences, and a short questionnaire followed, before he was presented with his final payoff. Figure 11 in Appendix B summarizes the experimental timeline for Player B.

Once Player B’s decisions had been recorded, Player A’s final payoff was calculated and all participants were paid on the day of participation.

Hypotheses and identification

Our design tests two main hypotheses, both grounded in the literature on responsibility avoidance through delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffell et al., 2016) and on responsibility attributions to algorithmic decision-makers (Feier et al., 2022; Kirchkamp and Strobel, 2019; Gogoll and Uhl, 2018).

H1 (Delegation): The introduction of potential punishment increases the likelihood that Player A delegates the decision to the algorithm.

H1 is grounded in empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. Bartling and Fischbacher (2012) also show that subjects are more likely to delegate morally consequential decisions even to a die roll, an early form of delegation to a non-intentional agent, suggesting that responsibility avoidance persists when the agent lacks intentionality. Whether the same logic extends to *algorithmic* delegates is less explored. Feier et al. (2022) report that delegation rates are similar for human and algorithmic intermediaries, suggesting that the kind of intermediary may matter less than the mere availability of one, but their design does not directly compare delegation rates across punishment regimes, leaving open the question H1 addresses. H1 is identified by the between-subject comparison of Player A’s delegation rate across the *Punishment* and *No-Punishment* conditions, which exploits the punishment manipulation as the only between-subject difference and holds fixed every other feature of the decision environment.

H2 (Punishment): Conditional on a low-payoff outcome for Player B, Player A is punished less when she delegated the decision to the algorithm than when she made it herself.

¹⁰The cost is fixed rather than proportional to the punishment level.

H2 mirrors H1 on the punishment side. [Bartling and Fischbacher \(2012\)](#) find that delegation to a randomization device still allows decision-makers to avoid some punishment, although less so than delegation to another human. [Feier et al. \(2022\)](#) find that decision-makers are rewarded less when a negative outcome is directly attributable to their own action, compared to when it results from an algorithm to which they delegated. However, our study differs from theirs in various respects: A different task, role separation between Player A and Player B, and the feature that their design does not control for relative performance at the individual level. As a result, in their setting the delegation decision carries a strong signal about Player A’s beliefs concerning her own ability relative to the algorithm. Thus, a failure to delegate may then be punished as overconfidence rather than for assuming responsibility, confounding inference on responsibility-shifting through pure intermediation with an inference about ability. H2 is identified by the within-subject comparison across the four (delegation x outcome) scenarios of Player B’s strategy-method punishment, which isolates the effect of delegation on punishment, holding fixed both the realized consequences for Player B and all between-subject heterogeneity in Player B’s punishment proclivity.

Two further design features deserve note. First, although the prediction task affects Player B’s payoff, the incentives of the two players are nominally aligned: Player A has no direct financial interest in Player B receiving the low payoff, and a successful prediction may weaken her own expected punishment.¹¹ This alignment mirrors canonical real-world settings for algorithmic delegation, in which the decision-maker has no direct material interest in the affected party’s outcome but bears reputational or moral consequences for outcomes. By replicating this alignment, our design speaks directly to policy-relevant cases. A standard intention-based punishment motive should therefore have less bite here than in dictator-game settings where the decision-maker can choose to be selfish. Second, although the underlying decision (a prediction under uncertainty) is not itself moral, the setup nonetheless qualifies as part of the *moral domain* as defined in the algorithm-aversion literature ([Gogoll and Uhl, 2018](#)): Player A’s choice imposes monetary consequences on a matched recipient, and a poor decision causes real monetary harm. [Comment: Think about where to best place this paragraph.]

4 Results

Data collection took place between February and June 2023 through the British online platform Prolific, with participation restricted to UK residents. A total of 322 subjects were recruited (161 assigned to Player A, 161 to Player B).¹² On average, Player A spent 12 minutes completing the experiment and earned £3.63, while Player B spent approximately 8 minutes and

¹¹In the *Punishment* condition, a better outcome for Player B should reduce expected punishment for Player A. In the *No-Punishment* condition there is no punishment to reduce. Either way, Player A’s incentives are weakly aligned with Player B’s.

¹²We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the *No-Punishment* condition. [Appendix D](#) reports the full ex-ante power calculations underlying this sample size and distribution.

earned £2.40. Treatment randomization was effective, as demonstrated by the balance across observable characteristics (Appendix A.1, Table 4).

In the following, we first test the two main hypotheses derived from our preregistered research questions. The subsequent analyses serve to understand the mechanism behind these results. They are exploratory in nature, though several of them were outlined as secondary analyses in the pre-registration.¹³

Player A – Delegation behavior

We begin by examining the delegation behavior of Player A. In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the decision to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 59.3% delegated. Figure 2 illustrates the difference. The introduction of potential punishment thus significantly *reduces* the likelihood of delegation to the algorithm (Pearson Chi-squared, $p = 0.033$; Fisher’s exact, $p = 0.041$ [Comment: drop Fisher exact test result?]), which reverses the direction posited by *Hypothesis 1*.¹⁴ [Comment: Those are the p-values for the two-sided test. In the design section, we have one-sided hypotheses. Should we instead have equal delegation as the hypothesis and then reject that? Particularly because the effect goes in the opposite direction as hypothesized, and because in the pre-registration only a non-directional research question was stated. Still, a direction was insinuated by the other hypothesis. In general, do stated and tested hypothesis have to coincide? Is it best practice to make the hypothesis rejectable, or do you just put the hypothesis in a way that it actually reflects hypothesized behavior?] Delegation shares around 50% are consistent with subjects, on average, finding the claim of equal relative performance between Player A and the algorithm conveyed by the design credible and their behavior reflects neither systematic preference for their own predictions nor systematic deference to the algorithm.

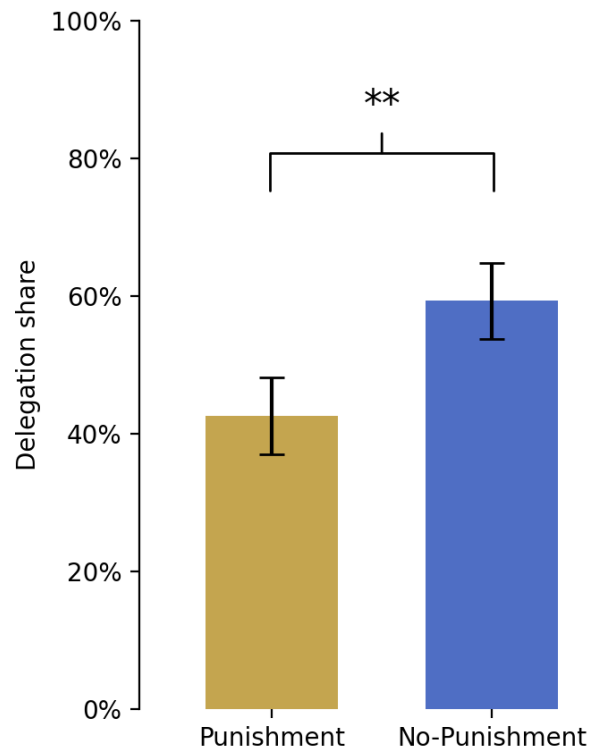
Result 1. *The possibility of punishment significantly reduces the likelihood that Player A delegates the decision to the algorithm.*

Next, we examine the robustness of the observed treatment difference in the delegation share. Specifically, we assess whether the difference in delegation rates remains when controlling for socio-demographic characteristics and Player A’s actual and perceived task performance. Ta-

¹³Unless stated otherwise, all tests are two-sided. We describe results as significant if $p < 0.05$ and as weakly significant if $0.05 \leq p < 0.10$. Because the delegation decision is a binary outcome compared between conditions, delegation shares are tested using Pearson chi-squared tests, with Fisher’s exact tests yielding the same conclusions throughout. Because punishment is elicited within-subject through the strategy method, punishment comparisons use paired *t*-tests, with Wilcoxon signed-rank tests as non-parametric counterparts. Regressions report robust (HC1) standard errors.

¹⁴A within-subject corroborator points the same direction. Player As in the *No-Punishment* condition were asked, after their actual delegation decision, the hypothetical question of whether they would have delegated had punishment been possible. Among the 81 respondents, 12 would have stopped delegating under hypothetical punishment (25% of delegators) while only 5 would have started (15.2% of non-delegators), constituting weakly significant switching behavior towards making the decision oneself with punishment possible (within-subject McNemar’s test $p = 0.090$), and yielding a hypothetical delegation rate of 50.6%.

Figure 2: Delegation shares by condition



Note: Bars show the share of Player As who delegated the prediction for Player B to the algorithm in each condition. The difference is significant at the 5% level under both a Pearson Chi-squared test ($p = 0.033$) and a Fisher's exact test ($p = 0.041$) [Comment: drop Fisher exact test result?].

ble 1 reports estimates from a Probit model of the delegation decision. Column (1) reports the unconditional treatment effect. **Comment: Perhaps revert and use a *Punishment* indicator.** Column (2) adds a vector of socio-demographic and other background characteristics. Columns (3) and (4) add Player A’s actual performance (number of correct predictions in the first ten rounds) and her weighted-average belief about own performance, respectively. Column (5) restricts the sample to subjects who passed the attention check.¹⁵ The treatment effect is robust across all five specifications: the coefficient on the No-Punishment indicator runs from 0.42 ($p = 0.034$) in the unconditional specification to 0.52 ($p = 0.025$) in the attention-check sample.¹⁶ Translating these into average marginal effects means that removing the punishment possibility raises the probability of delegation by between 16.5 and 19.2 percentage points across all specifications. Neither actual nor perceived performance is significantly associated with delegation in the pooled sample, although actual performance loads negatively and is weakly significant under the attention restriction ($p = 0.068$). We return to this pattern in discussing the mechanisms.

Player B – Punishment behavior

We now turn to Player B’s punishment behavior, beginning with the *Punishment* condition, in which Player B was given a real, payoff-relevant punishment choice. **Comment. Is this specification necessary?** Punishment is substantial across all four (delegation, outcome) scenarios of the strategy method (Figure 3).¹⁷ Despite the sizable punishment after *good* outcomes, Player Bs punish bad outcomes more than good outcomes, by a mean difference of £0.21 (significant at the 1% level), indicating that measured punishment carries real informational content rather than being random. Moreover, substantial punishment even after a *good* outcome need not reflect noise either, since Player Bs can form views about the appropriateness of delegation, independently of the realized outcome, and can punish accordingly.¹⁸

Holding the outcome fixed, Player A is punished no differently for delegating to the algorithm than for deciding herself. After a good outcome, average punishment is £0.329 for delegated versus £0.297 for self-made decisions. After a bad outcome, £0.482 versus £0.553. Thus, the punishment gap for bad outcomes runs in the predicted direction but is small and statistically insignificant (£0.071; paired t -test $p = 0.13$; Wilcoxon signed-rank $p = 0.54$). Regression analysis in Table 2 tells the same story: Punishment loads overwhelmingly on the outcome (bad-outcome coefficient 0.256, $p < 0.01$), the effect of delegation for good outcomes is essentially zero, and the *Delegated* $\times$ *Bad outcome* interaction – the differential that would capture any

¹⁵While the attention check was included on another screen, it may still correlate with who took the delegation decision seriously.

¹⁶Including a control for the random order in which the two delegation options appeared on screen leaves the treatment coefficient essentially unchanged (treatment coefficient 0.48 ($p = 0.019$), with the “algorithm first” order indicator at 0.27 ($p = 0.21$), starting from the specification in Column (3)).

¹⁷One-sample t -test: $p < 10^{-5}$ in every scenario.

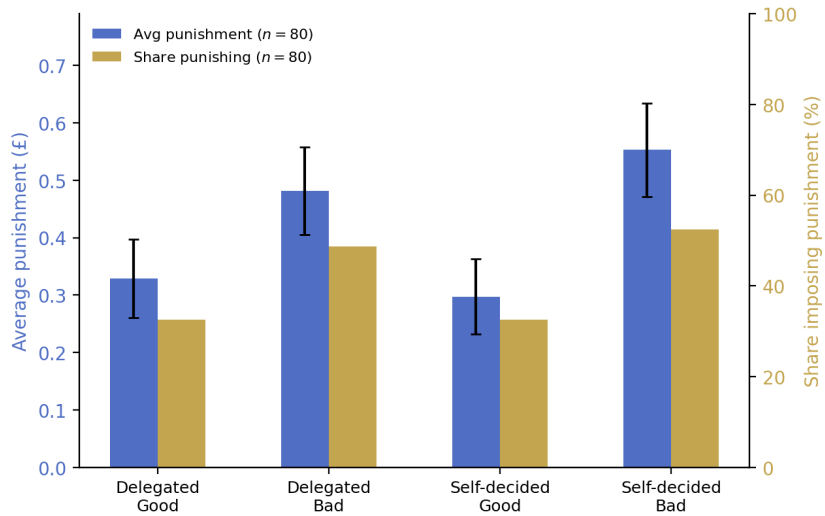
¹⁸For example, Coffman (2011) likewise documents non-trivial punishment in settings where the decision-maker’s action could not have been improved upon.

Table 1: Determinants of the delegation decision

	Dependent variable: <i>Delegation</i>				<i>Passers</i>
	<i>Full sample</i>				
	(1)	(2)	(3)	(4)	(5)
No-Punishment indicator	0.423** (0.199)	0.435** (0.204)	0.472** (0.206)	0.448** (0.205)	0.517** (0.230)
Task performance			−0.119 (0.079)		−0.158* (0.086)
Perceived performance				−0.072 (0.056)	
Age		−0.002 (0.009)	−0.002 (0.009)	−0.002 (0.009)	0.002 (0.010)
Female		0.175 (0.227)	0.222 (0.228)	0.212 (0.229)	0.407 (0.254)
Socio-economic status		−0.038 (0.071)	−0.035 (0.072)	−0.031 (0.071)	−0.082 (0.079)
Went to uni		0.036 (0.227)	0.059 (0.225)	0.025 (0.227)	0.124 (0.247)
Technology score		−0.069 (0.092)	−0.048 (0.095)	−0.062 (0.092)	0.014 (0.107)
Leadership position		0.404* (0.225)	0.422* (0.226)	0.399* (0.225)	0.253 (0.253)
Constant	−0.189 (0.141)	0.018 (0.537)	0.242 (0.570)	0.268 (0.580)	0.134 (0.645)
AME of No-Punishment (pp)	16.5** (<i>p</i> = 0.026)	16.6** (<i>p</i> = 0.026)	17.8** (<i>p</i> = 0.016)	17.0** (<i>p</i> = 0.022)	19.2** (<i>p</i> = 0.017)
N	161	159	159	159	130
Pseudo <i>R</i> ²	0.020	0.039	0.049	0.047	0.064

Note: Coefficients from a Probit regression of the delegation decision on the No-Punishment indicator and the controls listed in each column. Robust (HC1) standard errors in parentheses. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$ (two-sided). Two subjects are dropped in Columns (2)–(4) due to missing socio-demographic data. The *AME of No-Punishment* row reports the average marginal effect of the No-Punishment indicator on the probability of delegation, in percentage points.

Figure 3: Player B punishment by delegation and outcome scenario



Note: For each of the four (delegation, outcome) cells of the strategy method, the figure reports the average punishment chosen by Player B (left y -axis, in pounds; error bars are standard errors of the mean) and the share of Player Bs imposing non-zero punishment (right y -axis, in percent), in the *Punishment* condition (full sample, $n = 80$). The subsample of attention-check passers is shown in Appendix Figure 6. Hypothetical punishment in the *No-Punishment* condition is reported in Appendix Figure 8.

insulation from punishment for bad outcomes – is -0.103 and weakly significant ($p = 0.08$).¹⁹ The conclusion is unchanged when we restrict to subjects who passed the attention check, which was included on the punishment elicitation screen (Table 2, column 2).

Table 2: Determinants of Player B’s chosen punishment

	Dependent variable: <i>Punishment</i> (£)	
	<i>Full sample</i>	<i>Passers</i>
	(1)	(2)
Delegated	0.032 (0.052)	0.001 (0.054)
Bad outcome	0.256*** (0.071)	0.260*** (0.080)
Delegated $\times$ Bad outcome	-0.103^* (0.058)	-0.078 (0.052)
Player B belief about Player A perf.	-0.020 (0.032)	0.001 (0.037)
Age	0.002 (0.005)	-0.001 (0.006)
Female	-0.100 (0.128)	-0.064 (0.140)
Socio-economic status	0.083^* (0.047)	0.072 (0.049)
Went to uni	-0.196 (0.140)	-0.098 (0.142)
Technology score	0.073 (0.053)	0.030 (0.059)
Leadership position	0.083 (0.142)	0.061 (0.164)
Constant	-0.077 (0.328)	0.020 (0.374)
Observations	320	268
Subjects	80	67
Adj. R^2	0.077	0.030

Note: OLS regression of Player B’s chosen punishment levels. The sample is stacked over the four (delegation, outcome) scenarios of the strategy method (4 observations per Player B). The *Delegated* $\times$ *Bad outcome* interaction is the differential effect of delegation on punishment in the bad-outcome cell. Column (1) uses the full Punishment-condition sample; Column (2) restricts to Player Bs who passed the attention check on the punishment-elicitation screen. Standard errors clustered at the Player B level. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$ (two-sided).

[Comment: We could consider extending the analysis to explore determinants of punishment in each scenario or for the punishment difference between delegated and self-made decisions for either outcome.]

The same pattern holds on the extensive margin (Appendix Table 5). [FB: Say in a footnote that the corresponding regression results can be found in the appendix instead of including the reference just in parenthesis.] The share of Player Bs imposing *any* punishment is indistin-

¹⁹In the regression, because punishment for the different scenarios is elicited within-subject, any stable characteristic of Player B is differenced out and cannot, by construction, confound the within-subject comparison. Thus, the control variables merely capture the between-subject association with Player B’s *average* punishment level. We find no striking relationship between Player B’s average punishment level and either their socio-demographic characteristics or their beliefs about Player A’s *absolute* performance.

guishable between delegated and self-made decisions for each outcome (Figure 3, right axis). Punishment is also far from universal with 45.0% of Player Bs punishing in none of the four scenarios. Thus, within each scenario averages reflect a punishing minority rather than light, diffuse sanctioning.²⁰

Result 2. *Player A does not avoid punishment by delegating the decision to the algorithm.*

Comment: Perhaps phrase from the perspective of Player B? If so, align H2.

Thus, Player B neither punishes delegated decisions substantially less harshly, nor less frequently. Therefore, in our experimental setting Player A cannot avoid punishment by delegating to the algorithm, nor is she punished more when delegating the decision to the algorithm. Instead, punishment is determined almost entirely by the realized outcome. This result also bears on *Result 1* as any treatment difference in delegation frequencies driven by punishment avoidance would be inconsistent with Player B’s observed punishment behavior.

Mechanisms

Result 1 does not merely fail to support *Hypothesis 1*, but the effect is statistically significant in the opposite direction. The possibility of punishment discourages, rather than encourages, delegation to the algorithm. Thus, the prospect of punishment changes Player A’s delegation decision, even though we detect no punishment penalty for delegating. In the remainder of this section, we examine the data more closely and weigh candidate explanations that could reconcile the two main results.

Departures from canonical settings. *Hypothesis 1* was grounded in evidence from settings that differ from ours in ways that plausibly matter. In the modified dictator games of [Bartling and Fischbacher \(2012\)](#); [Coffman \(2011\)](#); [Hamman et al. \(2010\)](#); [Oexl and Grossman \(2013\)](#), the decision-maker chooses between a fair and an unfair allocation. Thus, her intent is more transparent, the chosen allocation is implemented with certainty, and the unfair option benefits her directly at the recipient’s expense. Punishment in these settings sanctions a deliberate, self-serving act, and delegation is attractive precisely because it blurs the link between the principal and that act. Our prediction task departs from these games in all three respects. The outcome is uncertain, so a bad outcome may reflect bad luck rather than bad intent, and neither option affects Player A’s own monetary payoff, so the incentives of the two players can be considered aligned, leaving no self-serving choice to police.²¹ With little intent worth sanctioning, any intention-based punishment motive loses its bite, and with it, the value to Player A of placing an intermediary between herself and the outcome. Nevertheless, we chose this task deliberately

²⁰ Among the 44 Player Bs who punish in at least one scenario, sanctions are mostly non-trivial, so the costly option is used as a substantive penalty rather than a minor gesture.

²¹ We consider the incentives aligned even though Player A’s payoff is only indirectly affected by her decision, as long as a better outcome for Player B weakly reduces Player A’s expected punishment in the *Punishment* condition.

since prediction under genuine uncertainty, by a decision-maker with no material stake in the affected party’s outcome, is the environment in which real-world algorithmic delegation typically operates.

This reasoning, however, explains why the predicted effect on delegation might vanish, not why it should reverse. The reversal calls for a motive that weighs against delegation when punishment is possible.

Anticipated differential punishment. The most natural candidate is anticipation of punishment. Player A may expect a delegated decision to be punished more harshly than a self-made one, and delegate less in response. Such an expectation would, for example, be consistent with evidence that observers view the delegation of morally consequential decisions to machines relatively critically (e.g., Gogoll and Uhl, 2018; Bigman and Gray, 2018; Jauernig et al., 2022; Bonnefon et al., 2024).

After the delegation decision but before learning the outcome, we elicited Player A’s beliefs about Player B’s punishment in each of the four (*delegation*, *outcome*) scenarios. We interpret the belief measure with caution for several reasons. First, the elicitation was unincentivized and took place after the delegation decision, so reports may reflect post-hoc rationalization rather than the beliefs that actually drove the choice. Second, hypotheticality may introduce additional noise as only two of the four scenarios remained relevant after Player A’s delegation decision in the *Punishment* condition, and all four were purely hypothetical in the *No-Punishment* condition. Anticipated punishment can affect behavior only where punishment is possible. We therefore focus in the analysis of punishment beliefs on the *Punishment* condition.

Two patterns emerge (Figure 4). First, Player As, on average, substantially overestimate punishment in every scenario. Second, Player As expect essentially the same punishment for delegated and self-made decisions, conditional on the outcome, correctly anticipating the pattern in Player B’s actual punishment behavior (Result 2).²² Thus, measured beliefs provide no support for a punishment anticipation mechanism for Result 1.²³

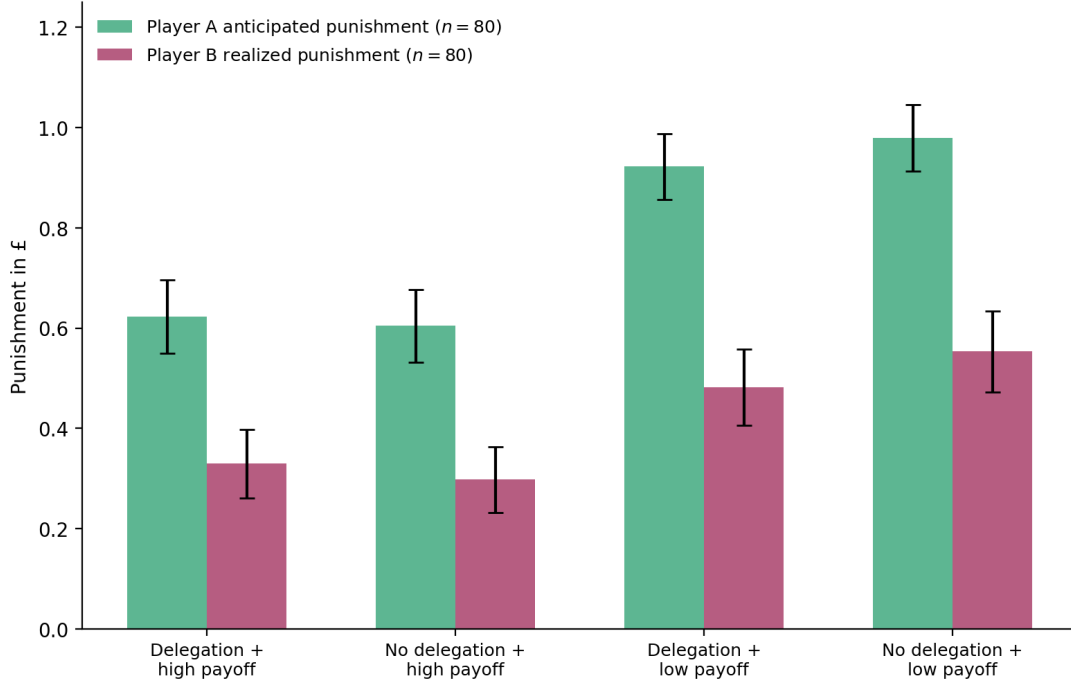
Even though the belief measure was not incentivized, and at the time of the belief elicitation, two of the four scenarios could not materialize anymore, punishment beliefs carry within-subject information. Table 3 reports a Probit regression of the delegation decision on Player A’s belief difference (*no delegation* – *delegation*) in expected punishment, restricted throughout to the *Punishment* condition.²⁴ Columns (1)–(3) use the simple average of the two belief differences (one for each outcome) as the single regressor: Column (1) reports the unconditional speci-

²²The expected punishment penalty for delegation is small and statistically indistinguishable from zero in both outcome states, and a within-subject test finds no asymmetry between the two.

²³The same holds for the fully hypothetical beliefs elicited in the *No-Punishment* condition, which closely track those in the *Punishment* condition (Appendix Figure 8). Thus, the manipulation appears not to have changed what subjects expected punishment to look like.

²⁴The two underlying belief measures compare expected punishment of a self-made and a delegated decision for each of the two possible outcomes for Player B, so that higher values correspond to relatively higher expected punishment for self-made decisions.

Figure 4: Realized vs. anticipated punishment, by scenario



Note: Player A's anticipated punishment and Player B's realized punishment, in pounds, across the four (*delegation, outcome*) scenarios in the *Punishment* condition (full sample). Whiskers denote ± 1 standard error. The subsample of attention-check passers is shown in Appendix Figure 7.

fication, Column (2) adds task-performance and socio-demographic controls, and Column (3) restricts the sample to subjects who passed the attention check on the belief-elicitation screen. Column (4) replaces the simple average with a weighted average, using Player A's elicited probability of producing the high payoff as the weight (the statistic that enters an expected-utility calculation). Column (5) enters the two outcome-conditional differences separately.²⁵

The estimated relationship is weak in the full sample but sharpens when restricting the sample to subjects who have passed the attention check on the belief elicitation screen. In Columns (1)–(2), expected differential punishment does not predict delegation. The average marginal effect is positive but not significant, at 1.3 to 1.4 percentage points per £0.10 of belief difference. Among attention-check passers, the effect more than triples and becomes significant. A £0.10 larger expected punishment penalty for deciding herself is associated with a 4.7 percentage-point higher probability that Player A delegates (Column (3), $p = 0.011$), and with 5.5 percentage points under the weighted specification (Column (4), $p = 0.006$). In the decomposition, the good-outcome belief difference carries the relationship (2.7 percentage points per £0.10, $p = 0.010$), while the bad-outcome difference is positive but noisier (1.7 percentage points, $p = 0.165$).

Comment: Consider including a discussion on the effect of performance (discussed later separately) and the observations about socio-demographics.

²⁵The two differences are nearly orthogonal in this sample (Pearson correlation -0.10 among passers), so the coefficients in Column (5) are additive partial slopes rather than competitors for the same variance. Under expected-utility reasoning, however, only their probability-weighted sum should enter the decision rule. We therefore treat Column (5) merely as a descriptive decomposition.

Table 3: Delegation and beliefs about punishment

	Dependent variable: <i>Delegation</i>				
	<i>Full Punishment</i>		<i>Punishment, passers</i>		
	(1)	(2)	(3)	(4)	(5)
Belief difference, average ($\bar{\Delta}$)	0.346 (0.407)	0.398 (0.448)	1.661** (0.700)		
Belief difference, weighted by Pr(good) (Δ^w)				1.992** (0.794)	
Belief difference, bad outcome (Δ^{bad})					0.590 (0.424)
Belief difference, good outcome (Δ^{good})					0.961** (0.412)
Task performance		-0.208* (0.111)	-0.362*** (0.136)	-0.342** (0.142)	-0.381*** (0.138)
Age		0.023* (0.012)	0.039** (0.016)	0.040** (0.016)	0.039** (0.016)
Female		0.152 (0.350)	0.950** (0.427)	1.026** (0.406)	0.845* (0.444)
Socio-economic status		-0.049 (0.111)	-0.006 (0.131)	-0.055 (0.133)	0.001 (0.126)
Went to uni		0.071 (0.363)	-0.278 (0.412)	-0.169 (0.408)	-0.275 (0.409)
Technology score		-0.061 (0.138)	0.048 (0.170)	0.072 (0.173)	0.033 (0.167)
Leadership position		0.302 (0.312)	-0.301 (0.408)	-0.279 (0.408)	-0.245 (0.411)
Constant	-0.198 (0.142)	-0.345 (0.826)	-1.228 (1.155)	-1.281 (1.147)	-1.115 (1.165)
AME of belief measure (pp per £0.10)	1.3 ($p = 0.388$)	1.4 ($p = 0.370$)	4.7** ($p = 0.011$)	5.5*** ($p = 0.006$)	
AME: good-outcome difference (pp per £0.10)					2.7** ($p = 0.010$)
AME: bad-outcome difference (pp per £0.10)					1.7 ($p = 0.165$)
Controls	×	✓	✓	✓	✓
Passers only	×	×	✓	✓	✓
N	80	78	60	60	60
Pseudo R^2	0.007	0.098	0.248	0.263	0.252

Note: Probit estimates with robust (HC1) standard errors in parentheses. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$ (two-sided). Sample restricted to the *Punishment* condition throughout. Belief differences are within-subject (*no delegation*) – (*delegation*) in expected punishment, so positive coefficients indicate that subjects who expect to be punished more harshly for not delegating (relative to delegating) are more likely to delegate. The *AME* rows report the average marginal effect of the belief measure entered in the respective column on the probability of delegation, in percentage points per £0.10 increase in the corresponding belief difference.

Taken together, the belief evidence is mixed. The measure itself is not pure noise. Within the *Punishment* condition, subjects whose beliefs embody a larger penalty for deciding themselves are more likely to delegate, so the delegation choice responds to belief asymmetries where they exist. But a margin whose average is zero provides no aggregate push. For anticipation of punishment to generate a 17 percentage-point delegation gap between treatments, Player As would need to expect systematically harsher punishment for delegated decisions. They do not and neither would such an expectation have been correct, as Player B’s realized punishment exhibits no delegation penalty either (Result 2). Anticipated differential punishment is thus a dimension along which subjects differ, but it cannot explain the treatment difference in delegation.

Result 3. *On average, Player A expects no punishment penalty for delegation, mirroring Player B’s actual punishment behavior. Within the Punishment condition, however, Player As who expect a larger penalty for deciding themselves are more likely to delegate.*

Selection into conditions. Nor is Result 1 an artifact of sample composition. Selection on observable characteristics cannot explain Result 1, as treatment randomization produced balance across socio-demographic and other background characteristics (Appendix A.1, Table 4). The same holds for performance. Average performance in the first ten rounds is balanced across conditions (2.76 vs. 3.09 correct predictions out of ten in the *Punishment* and *No-Punishment* conditions, respectively; t -test $p = 0.12$; Figure 5),²⁶ Player As in both conditions exhibit similar overconfidence about their own performance (4.65 vs. 4.87 in the *Punishment* and *No-Punishment* conditions, respectively; t -test $p = 0.47$),²⁷ and the treatment effect is robust to controlling for actual and perceived performance (Table 1, Columns (3)–(4)).

Performance and delegation. What differs across conditions is *who* delegates. A first indication already appears in the pooled delegation regression in Tables 1 and 3, where performance loads negatively and is weakly significant. This tendency crystallizes within the *Punishment* condition as the correlation between Player A’s score in the first ten rounds and the delegation decision is -0.19 , with $p = 0.092$ (Figure 5),²⁸ while a comparable relationship is weaker in the *No-Punishment* condition.²⁹ Framed at the level of treatment differences, the delegation gap widens with performance. The composition of delegators shifts accordingly. Player As who

²⁶To the extent this small raw difference matters at all, it works against Result 1, because the *Punishment* condition performs slightly worse, and as we will see, it is worse performers who delegate more in that condition, so any residual imbalance would attenuate rather than generate the treatment difference.

²⁷Confidence is measured by the weighted-average belief about own performance. This overconfidence may partly reflect the information that the algorithm performs equally well.

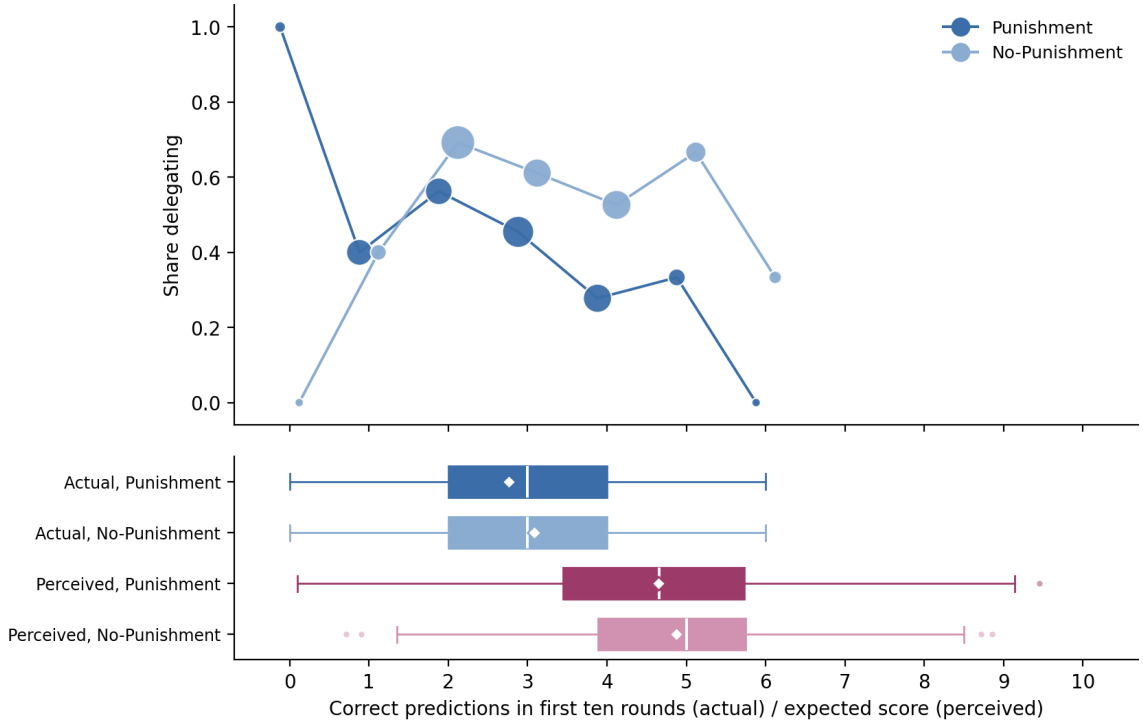
²⁸Among attention-check passers the relationship is even stronger ($r = -0.22$, $p = 0.083$).

²⁹A similar relationship holds for perceived performance in the *Punishment* condition. Player A’s weighted-average belief about her own performance enters with a small negative coefficient but is never statistically significant. Thus, actual performance discriminates sharply between delegators and non-delegators (2.38 vs. 3.00 correct out of ten, among attention-check passers) while perceived performance does not (4.69 vs. 4.71; Appendix Figure 9). Delegators under punishment are not subjects who *believe* they are worse than their algorithm, but they are subjects who *are* worse, and whose confidence does not reflect it.

delegated in the *Punishment* condition scored on average 2.47 out of ten in the first ten rounds, whereas delegators in the *No-Punishment* condition scored 3.06 ($p = 0.039$). The additional delegation in the *No-Punishment* condition thus comes disproportionately from the middle and upper part of the performance distribution. It is precisely these subjects who, when punishment is on the table, keep the decision for themselves.

Result 4. *In the Punishment condition, better-performing Player As are less likely to delegate. The additional delegation in the No-Punishment condition comes disproportionately from better performers.* Maybe leave out second sentence?

Figure 5: Delegation, performance, and confidence by condition



Note: The top panel shows the share of Player As at each first-ten-rounds score who delegated the final prediction, by condition. Dot size is proportional to the number of subjects at that score. The downward slope in the *Punishment* condition corresponds to the Pearson correlation of -0.19 reported in the text. The bottom panel shows the distributions of actual scores and of perceived scores (the weighted-average belief about own performance), by condition. Boxes span the interquartile range, the white line marks the median, and the white diamond the mean. Hue encodes the measure (blue for actual, magenta for perceived); shade encodes the condition (dark for *Punishment*, light for *No-Punishment*).

This relationship is interesting because its natural explanation, namely that worse performers delegate because they believe the algorithm outperforms them, is ruled out in our design. In [Feier et al. \(2022\)](#), the algorithm’s performance distribution is matched to the human distribution at the *population* level so that a high-performing subject who recognizes her position in that distribution rationally believes she outperforms the algorithm in expectation, and a low-performing one rationally believes the reverse. Delegation that declines with performance is then exactly what maximizing the expected outcome predicts. In our design, performance is equalized at the *individual* level and this is common knowledge. The pattern in the *Punishment*

condition must therefore reflect something other than beliefs about relative ability.³⁰

One reading of the relationship between performance and delegation could be a preference over *process*. Player As who have a good chance of producing the high payoff may prefer to bring it about themselves rather than through the algorithm. We refer to this as *process ownership* of the outcome. Equivalently, Player As with little chance of success may delegate to avoid being the proximate cause of a likely failure. Yet, while this motive can rationalize a negative relationship between performance and delegation, it would predict that relationship in *both* conditions and predict no treatment difference in delegation rates, unless the preference is *activated* by accountability, only becoming behaviorally consequential once Player B's stake in the decision is enforceable rather than merely material. Alternatively, Player A's preferences for delegation could vary with the amount of *uncertainty* inherent in a decision. Dietvorst and Bharti (2020) document that the preference for algorithms declines as the uncertainty inherent in a decision rises. **Comment: How does uncertainty in a task relate to difficulty of a task?** In our task, where most subjects score below five out of ten, better performance moves the success probability toward one half, arguably the point of maximal outcome uncertainty, so a distaste for delegating uncertain decisions would likewise depress delegation among better performers. However, also this channel applies to both conditions equally.

Effort to outperform the algorithm. Lastly, the prospect of punishment could motivate Player A to exert additional effort in making the decision for Player B. Despite the algorithm being calibrated to Player A's past performance, her final prediction is not yet made. Thus, a Player A may believe that she can try especially hard and raise the chance of a good outcome for Player B above the algorithm's, if the decision is not delegated. To the extent that Player A expects milder punishment after a good outcome, the prospect of punishment would then push her toward deciding herself. This possibility yields the prediction that non-delegators should improve their performance, specifically in the *Punishment* condition. Similar improvement in both conditions would instead indicate a treatment-independent (for instance, altruistic) motive, which cannot explain Result 1.

The data do not lend support to this story. Delegators do not make a final prediction themselves, so effort on the final prediction is observable merely for non-delegators. Measuring improvement as the difference between a subject's average absolute prediction error over the first ten rounds and her absolute error on the final prediction, non-delegators in the *No-Punishment* condition improve on their record, whereas non-delegators in the *Punishment* condition do not. Because all Player As faced the same images in the first 10 rounds and on the final prediction, this differential cannot reflect task-specific variation. Realized outcomes tell the same story with

³⁰We cannot rule out that some subjects did not fully internalize the individual-level calibration. Two observations limit this concern. First, any such misunderstanding should arise in both conditions alike, so it cannot explain why the relationship between performance and delegation emerges when punishment is possible, nor the treatment difference in delegation itself. Second, under this alternative, delegation should track perceived own performance, with more confident subjects keeping the decision. In the data, perceived performance is essentially identical for delegators and non-delegators.

45.5% of non-delegators in the *No-Punishment* condition earning the high payoff for Player B, compared with 21.7% in the *Punishment* condition, even though the two groups' performances are statistically indistinguishable over the first ten rounds (3.12 vs. 2.98 correct predictions; $p = 0.66$). If anything, effort seems to rise where punishment is impossible, which is the opposite of what a punishment-motivated effort story would predict.

Result 5. *The prospect of punishment does not induce additional effort. Non-delegators improve on their prior performance only where punishment is impossible.*

The signaling value of not delegating. Although non-delegators do not appear to exert additional effort, the delegation choice itself may communicate effort. By keeping the decision, Player A may want to convey to Player B that she takes the task seriously and intends to try hard on his behalf. Delegating, in contrast, could be perceived as shirking a decision that matters for another person. The signal lies in the choice itself and requires no actual additional effort, so Result 5 does not speak against this story. Result 3 seems to, if Player A had to expect greater punishment for shirking the effort to make a decision for Player B but, in the absence of punishment, can shirk without monetary consequences. Instead, if Player A expects no punishment penalty for delegation, there is little to gain from the signal. The two can be reconciled if Player A cares about Player B's judgment itself, beyond the punishment attached to it,³¹ or if our belief measure, unincentivized and elicited only after the delegation decision, captures the beliefs behind that decision imperfectly. A concern about being perceived as shirking would then reduce delegation where punishment is possible. In our environment, where no self-serving option exists and outcomes are noisy, the delegation choice is the only deliberate act Player B observes, which may lend this perception particular weight.

Comment: Maybe not make Result 5 explicit?

Comment: Instead of outcome, we can also compare time spent on the decision screen; Making the decision oneself in combination with common knowledge about equal performance in the first ten rounds may leave room for a signaling motive. By not delegating Player A may try to signal that they intend to try hard and beat their initial performance. Even though they do not seem to try, the signaling value stays. Explore this.

Taking stock. The reversal in Result 1 is robust, but its mechanism remains unresolved. It is not a composition artifact. The conditions are balanced in observables, performance, and confidence, and the treatment effect survives the corresponding controls. Anticipated differential punishment cannot explain the reversal (Result 3). Player A expects no punishment penalty for delegation on average, matching Player B's realized punishment (Result 2), even though be-

³¹Sanctioning power may make Player B's judgment consequential in Player A's eyes. A concern for how the delegation choice is perceived would then bind only when punishment is possible, even though the expected amount of punishment does not differ by delegation choice. Relatedly, such a concern on Player B's side could rationalize the positive punishment observed after good outcomes under delegation.

lie heterogeneity predicts delegation at the margin. Punishment-induced effort cannot explain it either, as improvement concentrates where punishment is impossible (Result 5). What the data document instead is that delegation is withheld under the possibility of punishment disproportionately by better-performing Player As (Result 4). Preferences over process ownership or over outcome uncertainty are consistent with this pattern, but treatment-independent and could explain Result 1 only if activated by accountability itself — an assumption our design cannot test, because the possibility of punishment and the presence of accountability coincide by construction.

5 Conclusion

tbd

Our findings add a potential determinant of algorithm use to the drivers documented in the aversion literature. Existing accounts locate the reluctance to rely on algorithms in the algorithm or the task, in doubts about ability, discomfort with opacity, or a taste for decisions made by hand. In our setting, the algorithm’s performance is equalized with the decision-maker’s by construction, and delegation nevertheless declines once the affected party can hold the decision-maker accountable. What appears to deter delegation is not what the algorithm does, but what delegating may say about the person who chooses it. Keeping the decision can signal engagement on behalf of the affected party, whereas handing it to the algorithm could be perceived as shirking. Accountability structures may thus shape algorithm use through perception concerns rather than through beliefs about performance. For the design of such structures, this cuts both ways. Holding human users accountable keeps them engaged with consequential decisions, in line with the intention of human-oversight requirements. Yet it may also discourage delegation in situations where the algorithm would decide just as well, and it does so disproportionately among the most capable decision-makers.

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Appendix

A Additional Tables

A.1 Condition balance — Player A

Table 4: Condition balance — Player A

Variable	<i>Punishment</i>	<i>No-Punishment</i>	<i>p-value</i>
Age	39.026	37.877	0.552
Female	0.487	0.494	1.000
Socio-economic status	5.179	5.321	0.564
Went to uni	0.575	0.556	0.928
Technology score	2.087	2.284	0.322
Leadership position	0.362	0.333	0.824

Note: Mean values by condition for Player A, with p -values from two-sided t -tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale; *Went to uni* indicates a university degree; *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items; *Leadership position* indicates self-reported management or supervisory experience.

Complementary to the individual treatment comparisons in Table 4, we can test their joint significance by regressing the treatment indicator on all characteristics in a linear probability model. The observables are jointly uninformative about treatment ($F(6, 152) = 0.36$, $p = 0.90$; $R^2 = 0.01$; $N = 159$). A logit specification yields the same conclusion (likelihood-ratio $\chi^2(6) = 2.26$, $p = 0.89$).

We do not separately report a balance table for Player B. Player B’s punishment behavior is identified within-subject across the four scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis.

A.2 Extensive margin of punishment

Table 5 re-estimates the punishment regression on the *extensive* margin, replacing the punishment amount with an indicator for whether Player B imposed any punishment in a given cell, estimated by probit. The conclusion matches the intensive-margin result in the body. A bad outcome sharply raises the probability of punishment—an average marginal effect of about 18 percentage points ($p < 0.001$)—whereas delegation leaves it essentially unchanged: the average marginal effect of delegation is -1.8 percentage points and statistically insignificant ($p = 0.46$; -2.9 percentage points among attention-check passers, $p = 0.29$). Player A is therefore no less *likely* to be punished when she delegates, just as she is punished no less *harshly*.

Table 5: Extensive margin of punishment — probit for $\Pr(\text{punishment} > 0)$

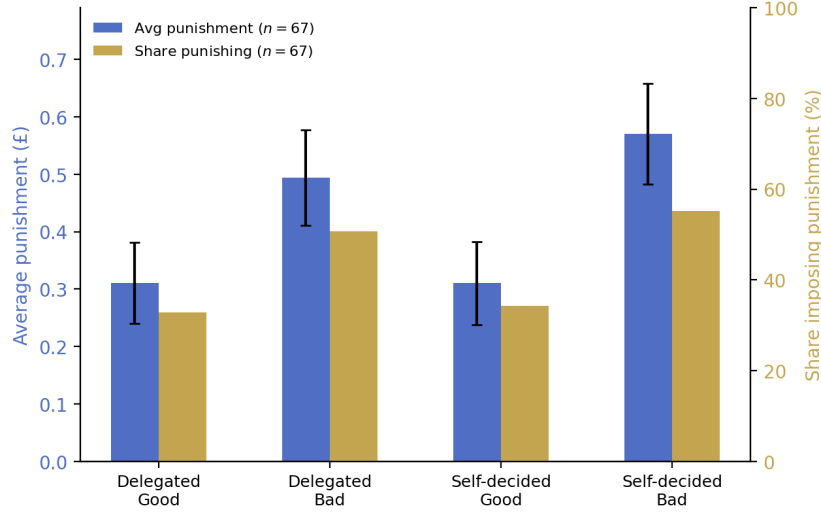
	Dependent variable: <i>Any punishment</i>	
	<i>Full sample</i>	<i>Passers</i>
	(1)	(2)
Delegated	0.002 (0.077)	−0.042 (0.078)
Bad outcome	0.556*** (0.139)	0.563*** (0.153)
Delegated × Bad outcome	−0.099 (0.095)	−0.072 (0.103)
Player B belief about Player A perf.	−0.083 (0.075)	−0.030 (0.082)
Age	−0.006 (0.013)	−0.013 (0.015)
Female	0.115 (0.272)	0.230 (0.291)
Socio-economic status	0.160 (0.126)	0.140 (0.129)
Went to uni	−0.827*** (0.320)	−0.706** (0.349)
Technology score	0.062 (0.118)	−0.023 (0.131)
Leadership position	0.062 (0.293)	−0.071 (0.321)
Constant	−0.382 (0.807)	−0.182 (0.864)
AME of Delegated (pp)	−1.8 ($p = 0.462$)	−2.9 ($p = 0.285$)
AME of Bad outcome (pp)	18.1*** ($p < 0.001$)	19.4*** ($p < 0.001$)
Observations	320	268
Subjects	80	67
Pseudo R^2	0.096	0.081

Note: Probit regression of an indicator for non-zero punishment ($1\{\text{punishment} > 0\}$) on a delegation indicator, a bad-outcome indicator, their interaction, Player B’s belief about Player A’s performance, and socio-demographic controls; the sample is stacked over the four (delegation, outcome) cells of the strategy method (4 observations per Player B). Column (1) uses the full Punishment-condition sample; Column (2) restricts to attention-check passers. Standard errors clustered at the Player B level. The *AME* rows report discrete average marginal effects on the probability of imposing any punishment, in percentage points (for Delegated, the change from $0 \rightarrow 1$ with the interaction moved consistently; for Bad outcome, likewise), computed by counterfactual prediction with p-values from a seeded 400-replication subject-cluster bootstrap. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$ (two-sided).

B Additional figures

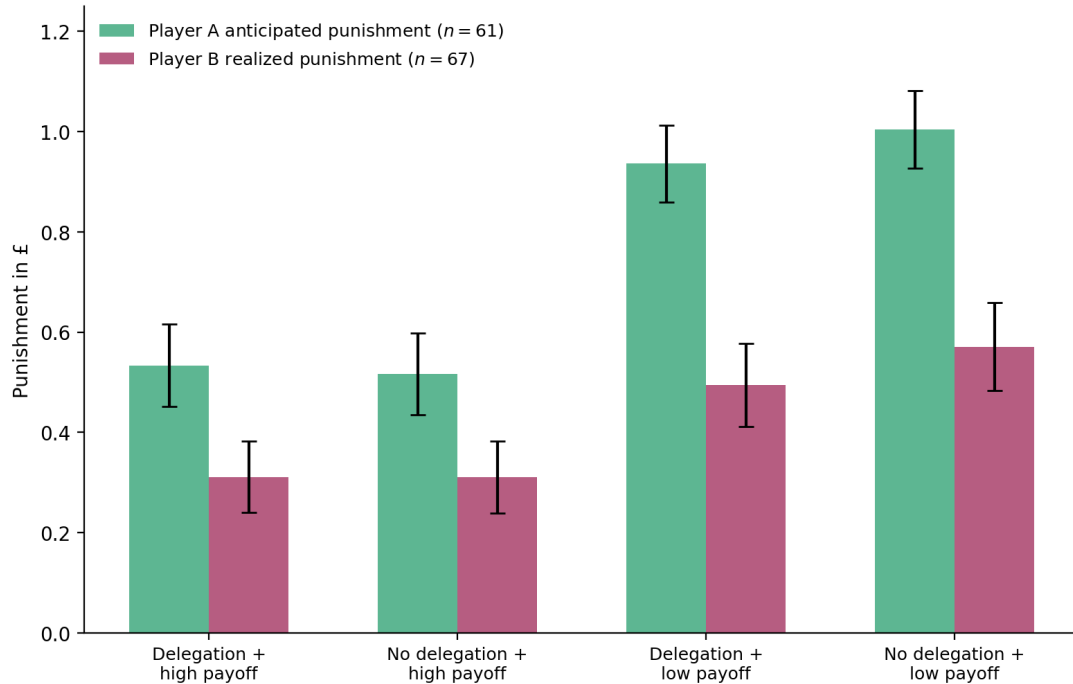
B.1 Punishment among attention-check passers

Figure 6: Player B punishment by delegation and outcome scenario, attention-check passers



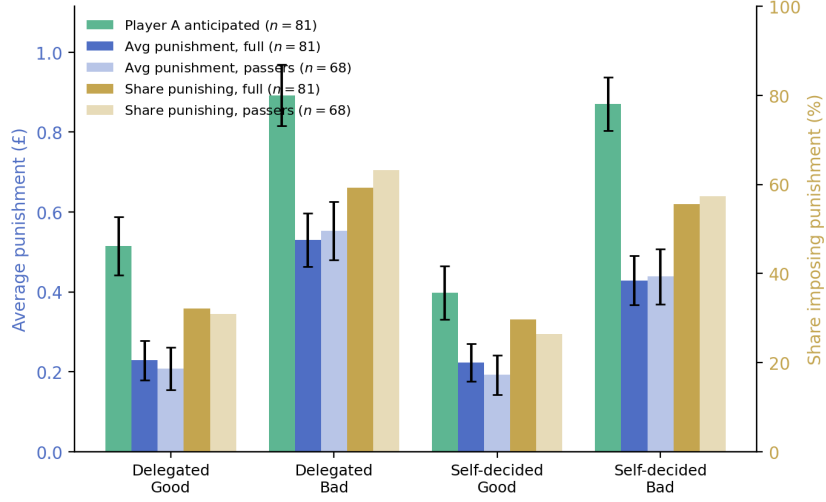
Note: As in Figure 3, but restricted to Player Bs who passed the attention check on the punishment-elicitation screen ($n = 67$). For each (delegation, outcome) cell of the strategy method, the figure reports the average punishment (left y -axis, in pounds; error bars are standard errors of the mean) and the share of Player Bs imposing non-zero punishment (right y -axis, in percent). The left-axis scale matches Figure 3.

Figure 7: Realized vs. anticipated punishment, attention-check passers



Note: As in Figure 4, but restricted to attention-check passers (Player A $n = 61$, Player B $n = 67$).

Figure 8: Hypothetical punishment by delegation and outcome scenario (*No-Punishment* condition)



Note: As in Figure 3, but for the *No-Punishment* condition (full sample $n = 81$, passers $n = 68$), in which Player Bs answered the same punishment questions but could not impose punishment in practice. For each (delegation, outcome) cell, the figure reports the average hypothetical punishment (left y -axis, in pounds; error bars are standard errors of the mean) and the share of Player Bs choosing non-zero punishment (right y -axis, in percent); the full sample is shown at full opacity and the attention-check passers overlaid at lower opacity. In addition, the green bars report Player A’s anticipated punishment in the *No-Punishment* condition (hypothetical, full sample, $n = 81$).

B.2 Hypothetical punishment

B.3 Performance and confidence by delegation choice

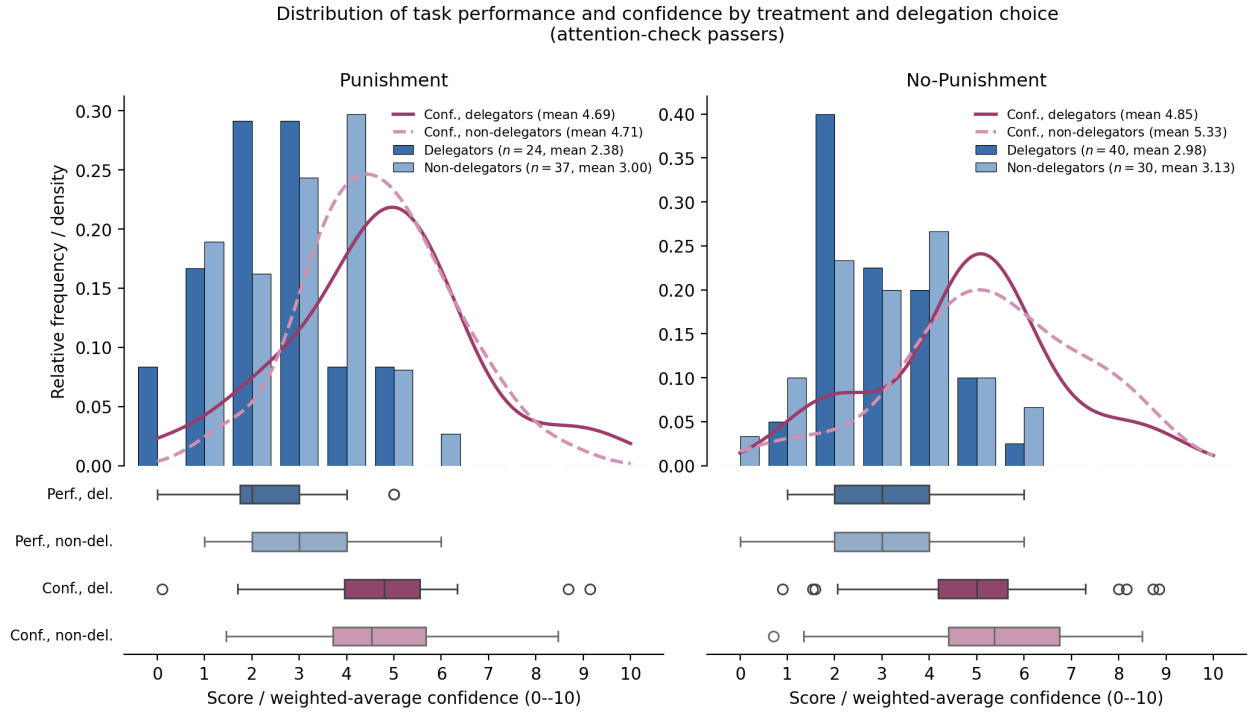
B.4 Experimental timelines

C Pre-registration

The study was pre-registered on AsPredicted under the title “Algorithms and Responsibility” (#133980) on May 30, 2023, after the collection of a pilot sample and before the collection of the main sample. The pre-registration is available at <https://aspredicted.org/hs9az8.pdf> and is reproduced in full on the following page.

The analyses in the paper deviate from this document in three respects. First, we recruited 322 participants instead of the planned 320. Attrition during the experiment required re-matching and produced one additional pair in the *No-Punishment* condition. Second, we compare delegation rates across conditions using a Pearson chi-squared test and Fisher’s exact test rather than the preregistered t -test. The delegation decision is binary, so these tests are the appropriate choice, and the conclusion is unchanged under either procedure. Third, the pre-registration specifies Mann-Whitney U and Kolmogorov-Smirnov tests for the punishment comparisons. Both are tests for independent samples, whereas punishment is elicited within-subject via the strategy method. We therefore report the Wilcoxon signed-rank test as the within-subject

Figure 9: Performance and confidence distributions by treatment and delegation choice



Note: Restricted to attention-check passers. Within each treatment, the top panel shows the share of Player As at each first-ten-rounds score, separately for those who delegated (dark blue) and those who did not (light blue), with the weighted-average performance belief overlaid as a KDE (delegators solid magenta, non-delegators dashed pink). Boxplot strips below the top panel report the same four series. In the *Punishment* condition, actual performance discriminates between delegators and non-delegators (means 2.38 vs. 3.00) while perceived performance does not (means 4.69 vs. 4.71). In the *No-Punishment* condition, neither actual nor perceived performance discriminates (actual: 2.98 vs. 3.13; perceived: 4.85 vs. 5.33). The additional delegators in *No-Punishment* therefore come disproportionately from the middle and upper portion of the actual performance distribution.

Figure 10: Experimental timeline — Player A

To be added.

Figure 11: Experimental timeline — Player B

To be added.

non-parametric counterpart alongside the parametric tests. Beyond these deviations, the pre-registration left open whether participants who failed the attention checks would be excluded. We report all analyses for the full sample in the main text and repeat them for the subsample of attention-check passers in the appendix.

Algorithms and Responsibility (#133980)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:
2023/05/30 16:27 (PT)

1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

We test whether:

RQ 1. punishment for a decision can be avoided if it is delegated to an algorithm (conditional on performance)

RQ 2. a punishment motive enters the delegation decision.

3) Describe the key dependent variable(s) specifying how they will be measured.

The experiment features a delegator and an evaluator. The delegator has to make a decision that is payoff-relevant only to the evaluator. The delegator may decide herself or delegate the decision to an algorithm, which is calibrated to match the delegators performance. Depending on the treatment, the evaluator may punish the delegator conditional on the outcome of the decision and the delegation decision (strategy method).

For RQ 1, we will look at evaluator punishment behavior and compare the amount and frequency of punishment between a delegated and non-delegated decision conditional on the outcome.

For RQ 2, we will look at the delegation frequency between the treatments (punishment possible vs punishment not possible).

4) How many and which conditions will participants be assigned to?

There are two conditions (punishment possible/no punishment possible) and two roles in each condition (delegator/evaluator).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

I will run parametric (t-test) and non-parametric (MWU, Kolmogorov-Smirnov) tests between punishment (amount and frequency) for delegated and non-delegated outcomes, conditional on the outcome. Similarly, I will run linear regressions with punishment behavior as the dependent variable and explore whether expected performance or any socio-demographic characteristics are predictive of differential punishment.

For delegators, I will again, run parametric (t-test) tests on delegation choices between treatments. Moreover, I will employ Probit regressions with the dependent variable being the delegation decision and a set of explanatory variables including treatment, beliefs about performance and socio-demographic characteristics.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

I will exclude people that access the experiment via mobile device. Moreover, I may exclude participants that failed the attention checks.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

I will recruit subjects via Prolific. I plan to recruit 320 participants, 160 per treatment, 80 per role per treatment.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

I already ran a pilot with 25 observations in the punishment treatment. I plan to include the data in the analysis.

Secondary analysis will include an investigation on how performance and beliefs about performance impact results. Moreover, I will look at the delegator's beliefs about expected punishment and their consistency with the delegation decision.

At this point, I already want to state that I may run another treatment in which the delegator may either delegate to another participant or an algorithm.

D Ex-ante power calculations

To be added